# Erick Ti

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## SUMMARY

As a full-stack and applied AI engineer at a high-velocity startup, I am focused on AI-powered tooling, LLM agents, large-scale data infrastructure, and shipping pragmatic systems end-to-end, from architecture to deployment and operations.

## EXPERIENCE

**Software Engineer** **May 2024 – Present**
*CelLink Corporation - San Carlos, CA*

- Built CelLink's company-wide AI agent (React/TypeScript, .NET 10, Anthropic SDK) for web and Microsoft Teams, used by all staff to query live business data and search internal docs, **cutting recurring software support requests by 20%**
- Engineered the AI agent's tools and guardrails: custom tool-calling orchestration, AST-validated read-only SQL queries, Qdrant document search, a second model pass that verifies answers against tool results
- Architected and maintained 10+ production ASP.NET MVC applications on SQL Server serving 500+ employees across the country with **99.9% uptime**, supporting validation of $12M+ in annual product revenue
- Optimized critical SQL Server stored procedures through cursor elimination, set-based CTE and CROSS APPLY rewrites, and composite indexing, **reducing p95 execution time from 11s to 500ms** in the company-wide product validation system
- Designed 7+ event-driven ETL pipelines (C# file-watcher services, AWS Timestream, MinIO) **processing 50,000+ manufacturing records daily**, then migrated them to a K3s Kubernetes cluster with multi-node failover and a 24-hour soak (stability) test
- Developed 9+ REST APIs for ERP integration and 30+ Grafana dashboards for real-time telemetry, improving operational workflow speed and **reducing manual data entry by 70%**

**Software Engineer / Database Intern** **Jan 2022 – Sep 2022**
*Ojibwe People's Dictionary - Minneapolis, MN*

- Refactored the Ruby on Rails backend and optimized MySQL queries for an Indigenous language preservation platform, **driving a 44% increase in organic monthly traffic** (24K to 34.6K users)

**Software Engineer Intern** **May 2022 – Aug 2022**
*Ecolab Inc. - Saint Paul, MN*

- Deployed 3 computer vision models on Azure Custom Vision + IoT Edge to detect packaging defects on a 4,000+ bottle/day production line, **achieving <1% false positive rate and 100% recall** with sub-second per-bottle inference
- Integrated PLC-driven reject logic with Azure Stream Analytics + Azure SQL and surfaced real-time results through Grafana

## PROJECTS

**Doppel - Machine Learning Music Recommendation Engine | doppel.erickti.com** **2026**

- Built a music recommendation engine that finds songs with a similar feel: pulls song candidates from Last.fm/ListenBrainz and reranks them by audio similarity with LAION-CLAP embeddings in Postgres + pgvector
- **Cut repeat-run latency ~60× (~12 min cold to ~12 s warm)** with a lazily grown embedding corpus, and benchmarked ranking quality with a 19-seed, 8-genre eval harness against cultural-retrieval and nearest-neighbor baselines
- Designed the Next.js frontend, with a deterministic SVG fingerprint for every track and a live ops panel showing pipeline telemetry

**Millennium - Self-Hosted Trading Card Portfolio Tracker | millennium.erickti.com** **2026**

- Built a confidence-scored import engine that matches collection CSVs against a **43,000+ printing catalog** to price the correct edition, then computes per-lot cost basis, market value, and unrealized gain/loss
- Shipped a Next.js UI with interactive valuation views and type-safe OpenAPI-generated TypeScript clients over a Django 5.2 + DRF API, backed by **873 pytest/Vitest/Playwright tests**
- Self-hosted on a Hetzner VPS: Caddy, containers, pull-based deploys, nightly systemd ETL, off-box backups to Cloudflare R2

## TECHNICAL SKILLS

**Languages:** TypeScript/JavaScript, Python, C#, SQL
**AI/LLM:** Anthropic SDK, MCP, tool-calling agents, RAG, evals, guardrails, Qdrant, pgvector, embeddings (Ollama, LAION-CLAP)
**Frontend:** React, Next.js, Vite, TanStack Query, Tailwind CSS | **Testing:** Vitest, Playwright, pytest, xUnit
**Backend:** Node.js/Express, FastAPI, Django + DRF, ASP.NET Core, EF Core
**Data & Infra:** PostgreSQL, Redis, SQL Server, Docker, Kubernetes (K3s), Vercel, Cloudflare R2, AWS (S3), GitHub Actions, GitLab CI

## EDUCATION

**University of Minnesota, Twin Cities** **Graduated: May 2023**
*Bachelor of Science in Computer Science | Minor in Information Technology | Minor in Digital Media Studies*